

Course COS 332 S1 2020
Test June 2020 Examination
Started 7/8/20 8:00 AM
Submitted 7/8/20 11:00 AM **LATE**
Due Date 7/8/20 11:00 AM
Status Completed
Attempt Score 72 out of 100 points
Time Elapsed 3 hours, 0 minute out of 3 hours
Instructions

- This is a three hour test to be written between 8:00 and 11:00.
- Note that the time allocated is longer than what would typically have been allocated, given the challenges of writing the test in a context where you may encounter connectivity problems and other technical challenges.
- This is a closed-book test. You are not allowed to obtain help from any person or other source while writing the test. You are, however, allowed to email the lecturer during the test at molivier@cs.up.ac.za if anything is unclear.
- All writing will be suspended at 11:00, even if you started writing less than three hours before that time.
- Note that the phrase "COS332 Hamming Code" is used to refer to Hamming code that uses even parity, and where the bits are numbered from right to left.

Results Submitted Answers
Displayed

• **Question 1**

0 out of 0 points

Do you promise not to consult any prohibited resources while writing this test?

Selected Answer: True

• **Question 2**

0 out of 1 points

For this and other multiple choice questions in the paper, select the alternative that fits the question best in each case.

The routing prefix in an IPv6 address (normally) consists of the ... of the address.

Selected Answer: First 32 bits

• **Question 3**

1 out of 1 points

The ASCII value of A is 41 (hex). Which of the following is a quoted-printable representation of "A=B"?

Selected Answer: =41=3D=42

• **Question 4**

1 out of 1 points

The software providing services on the internet is often known as a

Selected Answer: Daemon

- **Question 5**

1 out of 1 points

HTTP is used, amongst others, by Web browsers, WAP browsers, and applications that download maps and music, as well as many other applications. HTTP is on the

Selected Answer: Application layer

- **Question 6**

1 out of 1 points

Which of the following countries is not directly linked to SEACOM?

Selected Answer: DRC

- **Question 7**

0 out of 1 points

To terminate an FTP session, the following command(s) may be used:

Selected Answer: quit

- **Question 8**

0 out of 1 points

Which of the following is (are) bootstrapping protocols?

Selected Answer: DHCP

- **Question 9**

1 out of 1 points

The HTTP header contains a Host field in order to support

Selected Answer: Virtual hosting.

- **Question 10**

1 out of 1 points

To share a secret between n persons so that at least k of them have to cooperate to determine the secret, one may use the method of

Selected Answer: Shamir

- **Question 11**

0 out of 1 points

Which aspect is not standardised by RS-232?

Selected Answer: Functional

- **Question 12**

1 out of 1 points

If the receiver wants to react positively to an RS-232 RTS request, it will send a ...

Selected Answer: CTS

- **Question 13**

0 out of 1 points

A null modem is

Selected Answer: A 'Modem' used between a digital device and a digital medium.

- **Question 14**

1 out of 1 points

The entry 0.0.0.0/0 in a routing table is used to provide information about the

Selected Answer: Default gateway.

- **Question 15**

0 out of 1 points

Which transport layer protocol(s) is/are used by DNS?

Selected Answer: UDP

- **Question 16**

0 out of 1 points

Given that the speed of light is 3×10^8 m/s, how long does it take a signal to travel from an earth station to a geostationary satellite and back to another earth station?

Selected Answer: 0,12s

- **Question 17**

1 out of 1 points

In MACA, RTS is an acronym for

Selected Answer: Request To Send

- **Question 18**

1 out of 1 points

The TTL field in an IP datagram

Selected Answer: Determines the maximum number of hops over which the packet can be forwarded.

- **Question 19**

1 out of 1 points

A zone transfer refers to the process of

Selected Answer: Copying DNS entries from a primary server to a secondary server.

- **Question 20**

1 out of 1 points

Suppose the first octet of a UTF-8 character is 193. How long (in octets) will the entire character be?

Selected Answer: 2

- **Question 21**

0 out of 1 points

Hop count refers to

Selected Answer: The number of times a packet has been bounced between two routers.

- **Question 22**

0 out of 1 points

When TCP uses a three way handshake, the first ACK is for ... data byte.

Selected Answer: Whatever the first message included

- **Question 23**

1 out of 1 points

A modem

Selected Answer: Modulates a digital signal so that it can be transmitted via an analog medium over a long distance.

- **Question 24**

0 out of 1 points

Which field does not occur in the UDP header?

Selected Answer: Length

- **Question 25**

0 out of 1 points

TFTP

Selected Answer: More than one of the above

- **Question 26**

0 out of 1 points

Which of the following is not a valid ASN.1 constructor?

Selected Answer: CHOICE

- **Question 27**

1 out of 1 points

Two stations are communicating with one another using HDLC. Station A has already sent 15 frames of information to station B and station B has sent 9 frames of information to station A. All transmitted frames have been received. Station B now sends a further frame to station A. What are the values of N(R) and N(S) in this frame?

Selected Answer: N(R)=15, N(S)=9

- **Question 28**

0 out of 1 points

Free Space Optics (FSO) is a technology that

Selected Answer: More than one of the above

- **Question 29**

0 out of 1 points

Consider the problem of transparency in SMTP. Which of the following is true?

Selected Answer: In most e-mail programs users should avoid typing a full stop on a line of its own, because that will prematurely terminate the body of the message and cause only the part prior to the full stop to be sent.

- **Question 30**

0 out of 1 points

The amount of white space used in a command or data to be sent across a network may or may not be important. For example, in a command such as GET /

... the protocol may prescribe that only a single space character should occur between the GET and the /, or that any number of spaces may be used. Which of the following message description languages allow(s) one to prescribe the number of space characters that may occur in such a message?

Selected Answer: ABNF

- **Question 31**

1 out of 1 points

An application protocol will typically respond with a message in the ...-range when it successfully completed a request.

Selected Answer: 200s

- **Question 32**

1 out of 1 points

SAT3 is a

Selected Answer: Undersea cable

- **Question 33**

1 out of 1 points

Which well-known port is reserved for HTTPS?

Selected Answer: 443

- **Question 34**

0 out of 1 points

ADSL

Selected Answer: Is used between the exchange and the user of the telephone service.

- **Question 35**

1 out of 1 points

Two stations are communicating with one another using HDLC. Station A has already sent 15 frames of information to station B and station B has sent 9 frames of information to station A. All transmitted frames have been received. Station B now sends a further frame to station A. What are the values of N(R) and N(S) in this frame?

Selected Answer: N(R)=7 , N(S)=1

- **Question 36**

1 out of 1 points

Let S be the source address of an IP datagram. Let D be the destination address of an IP datagram. Let N be the netmask used on a router. Let & represent the logical and operation. If $S \& N = D \& N$ then the router will

Selected Answer: Directly deliver the packet.

- **Question 37**

1 out of 1 points

If an HTTP server has insufficient storage available to process a request, it should return the following status code:

Selected Answer: 507

- **Question 38**

0 out of 1 points

Assume you are writing a server that is intended to be accessed by a Telnet client.

Selected Answer: The server should issue an ANSI escape sequence to disable local echoing at the client; the server should then echo everything received.

- **Question 39**

1 out of 1 points

The UTF-8 representation of U+05D0 is

Selected Answer: D7 90

- **Question 40**

0 out of 1 points

On which ISO OSI layer(s) does the World Wide Web fit?

Selected Answer: 6

- **Question 41**

0 out of 1 points

Latency refers to

Selected Answer: The maximum possible speed at which a network can operate.

- **Question 42**

0 out of 1 points

Which of the following is the latest ratified (accepted) standard for WLANs?

Selected Answer: IEEE 802.11g

- **Question 43**

0 out of 1 points

An application protocol will typically respond with a message in the ...-range when security or configuration errors prevented it from completing a request.

Selected Answer: 400s

- **Question 44**

0 out of 1 points

If SMTP cannot deliver a message, it may

Selected Answer: Send the message to a secondary server.

- **Question 45**

1 out of 1 points

Peer in ISO OSI terminology refers to

Selected Answer: Software on the same layer as other software

- **Question 46**

1 out of 1 points

If a computer wants to send data to a modem, lines are used in the following order:

Selected Answer: DTR, DSR, RTS, CTS, TD

- **Question 47**

1 out of 1 points

In data communications a protocol is always ...

Selected Answer: A detailed description of the rules to be followed when communicating.

- **Question 48**

0 out of 1 points

The most important use(s) of RFCs is (are)

Selected Answer: All of the above

- **Question 49**

1 out of 1 points

An email message that contains a PDF document as an attachment may use the following MIME subtype to encapsulate the mail message and its attachment.

Selected Answer: multipart/mixed

- **Question 50**

1 out of 1 points

To use POP3 anonymously, one uses the command

Selected Answer: None of the above

- **Question 51**

0 out of 1 points

X.509 is a standard of the

Selected Answer: IEEE

- **Question 52**

1 out of 1 points

Fewer collisions occur with the slotted Aloha protocol, than with pure Aloha, since

Selected Answer: Slotted Aloha reduces the number of frames that overlap partially.

- **Question 53**

1 out of 1 points

Which of the following is not a valid UTF-8 sequence?

Selected Answer: F2 B3 7F BA

- **Question 54**

1 out of 1 points

The advantage of a crossbar switch over a bus is the fact that it

Selected Answer: None of the above

- **Question 55**

0 out of 1 points

Ethernet addresses have to be unique. This is accomplished as follows:

Selected Answer: The network administrator assigns a unique address to each card.

- **Question 56**

0 out of 1 points

Multicast (class D) IP addresses may be used to establish

Selected Answer: Point-to-multipoint TCP connections.

- **Question 57**

1 out of 1 points

The theoretical upper time bound in a CSMA/CD protocol to send frame n is ... slots.

Selected Answer: None of the above

- **Question 58**

1 out of 1 points

The layer 2 data delineation problem is about

Selected Answer: Marking the boundaries of a frame.

- **Question 59**

1 out of 1 points

The structure used to store data used by SNMP is known as ...

Selected Answer: MIB

- **Question 60**

0 out of 1 points

When a token is lost in an IEEE 802.5 network, the network uses the following algorithm to regenerate the token:

Selected Answer: Bully

- **Question 61**

1 out of 1 points

Which of the following is not an SNMP request?

Selected Answer: Reboot

- **Question 62**

1 out of 1 points

You receive the bit string that follows encoded as a COS332 Hamming value. At most one bit could have changed. Which bit changed? Write 0 if no bit changed during transmission. Received data: 111010010111

Selected Answer: 0

- **Question 63**

1 out of 1 points

You receive the bit string that follows encoded as a COS332 Hamming value. What 12-bit string of bits was originally transmitted if at most one bit could have changed? Received data: 110110011011

Selected Answer: 110110010011

- **Question 64**

1 out of 1 points

You receive the bit string that follows encoded as a COS332 Hamming value. At most one bit could have changed. Which bit changed? Write 0 if no bit changed during transmission. Received data: 111011111001

Selected Answer: 4

- **Question 65**

1 out of 1 points

You receive the bit string that follows encoded as a COS332 Hamming value. What 12-bit string of bits was originally transmitted if at most one bit could have changed? Received data: 101101111100

Selected Answer: 001101111100

- **Question 66**

1 out of 1 points

You receive the bit string that follows encoded as a COS332 Hamming value. At most one bit could have changed. Which bit changed? Write 0 if no bit changed during transmission. Received data: 101010001011

Selected Answer: 9

- **Question 67**

1 out of 1 points

You receive the bit string that follows encoded as a COS332 Hamming value. At most one bit could have changed. Which bit changed? Write 0 if no bit changed during transmission. Received data: 101110010100

Selected Answer: 1

- **Question 68**

2 out of 2 points

You receive the bit string that follows encoded as a COS332 Hamming value. At most one bit could have changed. What decimal number was supposed to have been sent (prior to any encoding)? Received data: 010010000010

Selected Answer: 64

- **Question 69**

2 out of 2 points

You receive the bit string that follows encoded as a COS332 Hamming value. At most one bit could have changed. What decimal number was supposed to have been sent (prior to any encoding)? Received data: 100110010100

Selected Answer: 211

- **Question 70**

2 out of 2 points

Convert the byte (sequence) EE AA B2 from UTF-8 to Unicode. Omit the U+ Provide your answer using four hexadecimal digits.

Selected Answer: EAB2

- **Question 71**

2 out of 2 points

Convert U+8F18 to UTF-8. The answer should consist of a string of 2-digit bytes separated by a single space between bytes.

Selected Answer: E8 BC 98

- **Question 72**

2 out of 2 points

Convert the byte (sequence) EE A5 B2 from UTF-8 to Unicode. Omit the U+ Provide your answer using four hexadecimal digits.

Selected Answer: E972

- **Question 73**

2 out of 2 points

Convert the byte (sequence) EE B4 87 from UTF-8 to Unicode. Omit the U+
Provide your answer using four hexadecimal digits.

Selected Answer: ED07

- **Question 74**

2 out of 2 points

Convert U+4EE8 to UTF-8. The answer should consist of a string of 2-digit bytes separated by a single space between bytes.

Selected Answer: E4 BB A8

- **Question 75**

1 out of 1 points

Assume processes A and B are communicating using TCP. For the sake of this question assume that all actions are atomic; when a message is sent, it is received essentially immediately at the other end. Stated differently, two events will never overlap, with one exception - see events 3 and 4 below. Assume A has just sent a message to B with SEQ=100, LEN=500, ACK=1000 and WIN=2000. Immediately after that B sent a message to A with LEN=300 and WIN=1000. Then the following events happen in sequence outlined below. However, events 3 and 4 happen at exactly the same moment and these messages both arrive at their destinations a split second later.

A sends a message to B with LEN=200.

B consumes 500 bytes from its buffer.

A sends a message to B with LEN=600.

B sends a message to A with LEN=200.

A consumes 1000 bytes from its buffer.

A sends a message to B with LEN=300

Below events 1, 3, 4 and 6 will be referred to as messages 1, 3, 4 and 6, respectively.

What is the SEQ value sent along with message 1?

Selected Answer: 600

- **Question 76**

1 out of 1 points

What is the window advertisement value sent along with message 4?

Selected Answer: 1300

- **Question 77**

1 out of 1 points

What is the ACK value sent along with message 3?

Selected Answer: 1300

- **Question 78**

1 out of 1 points

What is the ACK value sent along with message 4?

Selected Answer: 800

- **Question 79**

1 out of 1 points

What is the ACK value sent along with message 1?

Selected Answer: 1300

- **Question 80**

1 out of 1 points

The start of frame delimiter follows the preamble in an Ethernet frame. Use a sequence of zeroes and ones to indicate the value of this start of frame delimiter. Do not insert any spaces into your answer.

Selected Answer: 10101011

- **Question 81**

1 out of 1 points

The process of insertion of special values into SDLC/HDLC messages to ensure that data is not accidentally deemed to be delimiting characters is known as ____.

Selected Answer: bit stuffing

- **Question 82**

1 out of 1 points

The start of a token ring message is indicated by the octet JK0JK000.

Suppose the character ^ indicates a "high" signal and "_" (the underscore) indicates a "low" signal, the what will the actual signal look like that represents this octet. Assume the this octet starts with a "low" signal. (Include this "low" signal in your answer to account for all the signal levels used to represent this octet; do not insert any spaces in your answer.)

Selected Answer: __^^^__^^^_^_^_

- **Question 83**

1 out of 1 points

The payload of an Ethernet frame cannot exceed ____ octets.

Selected Answer: 1500

- **Question 84**

1 out of 1 points

The bit pattern 01111110 used in SDLC/HDLC to delineate frames is known as a ____.

Selected Answer: flag

- **Question 85**

1 out of 1 points

Your company uses a classless IPv4 address for which 14 bits are used as the network portion. One of the addresses on your company's network happens to be

99.33.22.13 . Use this scenario for all subnetting questions that follow. What is the network address of your company (in CIDR notation)?

Selected Answer: 99.32.0.0/14

- **Question 86**

1 out of 1 points

What is the broadcast address of your company? (Do not use CIDR notation for such specific addresses.)

Selected Answer: 99.35.255.255

- **Question 87**

1 out of 1 points

You are requested to subnet this address such that it can handle 10 subnets, with 200 hosts on each. What netmask will you use inside the organisation?

Selected Answer: 255.255.192.0

- **Question 88**

1 out of 1 points

What is the network address of subnet 1? Write your answer in CIDR notation.

Selected Answer: 99.32.64.0/18

- **Question 89**

1 out of 1 points

What is the broadcast address of subnet 12? (Do not use CIDR notation.)

Selected Answer: 99.35.63.255

- **Question 90**

0 out of 1 points

What is the network address of subnet 4? Write your answer in CIDR notation.

Selected Answer: 99.32.128.0/18

- **Question 91**

2 out of 2 points

On which subnet is the address 99.32.134.178 located?

Selected Answer: 2

- **Question 92**

2 out of 2 points

On which subnet is the address 99.34.20.77 located?

Selected Answer: 8

- **Question 93**

0 out of 0 points

Have you cheated while writing this test?

Selected Answer: True

Sunday, April 11, 2021 6:39:27 PM SAST